Tanuj Palaspagar

Software Developer Engineer

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EXPERIENCE

Zennith Technologies Inc. Mar 2025 - Present

Software Engineer

- Automated secret scanning, vulnerability scans and build signing in the Jenkins pipeline every release uses; 63 high/critical findings caught.
- Shipped the Kafka consumer that records each client payment exactly once, even when servers crash; no duplicates found in ~120M events.
- Owned the Python SDK clients use to send events; rebuilt it on aiokafka for 2x the throughput without clients changing any of their code.
- Traced 186 false missing-payout alerts to a timezone bug and fixed it that day, then added a CI rule failing any Airflow job not in UTC.
- Built a dispute report on the lineage API that traces any payment end to end; a "were we paid twice?" answer went from 20 minutes to 5.

HP Inc. Jun 2024 - Mar 2025

System Software Engineer

- Migrated 6 of 11 Workforce Experience Platform services from Jenkins to OneCloud's Azure DevOps, gated on Trivy, Veracode and SonarQube.
- Resolved ~140 SonarQube issues in the Java Spring Boot AWS Marketplace service, 21 critical, incl. replacing CRC checksums with SHA.
- Restored login across environments by raising a request-header limit to 20KB, then pinned releases when a redeploy undid it 9 days later.
- Built the Pro and Elite subscription plans with feature-flag entitlements, then traced their missing EU rollout to an S3 403 in a sync Lambda.
- Created a fastn.ai Slack connector, Maestro sync pipelines for 5 services, and mapped the Gekko and GraphQL store APIs.

HP Inc. May 2023 - Aug 2023

Software Developer Intern

- Redesigned HP TechPulse reporting into a flexible, configurable analytics solution for enterprise IT teams managing device fleets.
- Engineered a secure device-telemetry pipeline on AWS Kinesis and Redshift, streaming enterprise fleet data into the new reporting layer.
- Cut on-device telemetry latency from 60ms to under 17ms by moving the Windows client onto WMI event subscriptions.
- Integrated the AWS QuickSight APIs to turn TechPulse telemetry into scalable dashboards that enterprise IT teams could act on.
- Customized TechPulse analytics to the needs of each client fleet, reducing operational overhead for enterprise IT teams.

EDUCATION

The University of Texas at Arlington, *Master of Science in Computer Science* May 2024

University of Mumbai, *Bachelor of Engineering in Computer Engineering* May 2022

RESEARCH & PROJECTS

AI Sports Commentary Engine | Fayble Inc. Nov 2025 - Mar 2026

Python, Apache Kafka, Gemma 3, Dia TTS, LTX-2, ComfyUI, RTX 3090

- Designed an event-driven Kafka commentary engine from an ambiguous early brief, joining schedule and play topics into narrated video.
- Reduced Gemma 3 27B from 81s of multi-option output to one grounded ~12s take per event with prompt design, token caps and cleanup.
- Synthesized each line with Dia TTS from LLM emotion tags, padding the audio sample-exact to 375-frame video slots for frame-accurate sync.
- Ran the 19B LTX-2 video diffusion model on a 24 GB RTX 3090 with distilled FP8 weights, rendering 1280x704 clips in 8 denoising steps.
- Defined live-inference GPU requirements with the founder, estimating Dia's real-time throughput from measured RTX 3090 memory bandwidth.

Heimdall - Telemetry Character-Device Driver | github.com/tanujp99/heimdall Aug 2026 - Present

C, Linux Kernel Modules, ioctl, sysfs, Kbuild, QEMU, GitHub Actions

- Wrote an out-of-tree Linux char-device driver in C serving /dev/telemetry0 via read, ioctl and sysfs behind a versioned source_ops seam.
- Built three swappable sources behind the seam, including a timer feeding a spinlock-guarded ring buffer and a byte-deterministic replay source.
- Verified it with a 23-check QEMU suite and CI across kernels 5.15 and 6.8 plus an aarch64 cross-build, asserting a clean kernel log each run.

Credit Card Fraud Detection Sep 2021 - Feb 2022

Logistic Regression, Random Forest, KNN, Support Vector Machine, Python

- Benchmarked four classifiers on 284,807 Kaggle card transactions with a 0.172% fraud rate, comparing precision, recall and F1 for each model.
- Countered the class imbalance by undersampling genuine transactions, then evaluated every model on sampled and unsampled data.
- First author to international publication: Study of Machine Learning Algorithms for Credit Card Fraud Detection (IRJMETS Vol. 4, 2022)

SKILLS

Programming Languages: Python, Java 17, C, Go, SQL, GraphQL, Bash

Frameworks & Tools: Spring Boot, Flask, FastAPI, Kafka, PySpark, Airflow, Jenkins, Docker, Kubernetes, Azure DevOps, Maven, SonarQube

Cloud & Platforms: AWS (S3, DynamoDB, Secrets Manager, Kinesis, RedShift, QuickSight), Hugging Face, QEMU, Linux (Ubuntu/Arch)